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Nicholas Larsen

AI/ML Engineer

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Education

Master of Science in Computer Science, Dallas, TX (4.0) **Concentration:** AI/ML
Southern Methodist University December 2022
Bachelor of Science in Computer Science (3.670) **Minor:** Mathematics
Texas Christian University, Fort Worth, TX May 2018

Technical Experience

AI/ML Software Engineer **December 2024 - Present**
Kubota *Grapevine, TX*

- Develops real-time computer vision software that interfaces with Luxonis and ZED X cameras and Ouster lidar, feeding perception data to autonomous vehicle systems for both guidance and safety functions.
- Develops sensor fusion that detects objects in RGB images and localizes them in 3D using lidar.
- Owns sensor calibration for the vehicle's camera and lidar platforms.
- Trains and evaluates computer vision models for construction and agriculture automation, owning the pipeline from data collection and labeling through deployment.
- Optimizes and deploys models with SNPE on Qualcomm-based Luxonis cameras and TensorRT on Nvidia Jetson.
- Drives technical decisions for perception and autonomy components and their vehicle-level integration.
- Builds proof-of-concept demos and documentation for technical and executive audiences, validating feasibility and maturing Physical AI capabilities toward commercialization.

Software Engineer Senior **May 2022 - December 2024**
Lockheed Martin *Fort Worth, TX*

- Architected Catia V5 and 3DX environments with proprietary software, 3rd party applications, corporate settings, and other enhancements giving Lockheed designers an edge on the competition.
- Overhauled manual deployment of Catia V5 and 3DX CAA customizations with GitLab pipelines, delivering CI/CD for 50+ CAD/CAM applications to air-gapped and unclassified networks.
- Managed coding standards and approved code reviews for custom 3DX and V5 applications.

Software Engineer **July 2018 - May 2022**
Lockheed Martin *Fort Worth, TX*

- Developed an application for releasing design engineer's drawings and 3-D models. Leveraged an Oracle database for storing the CAD Data, meta-data parts and positions on different airplanes, and release state of drawings.
- Designed new features for engineers to create retro-fit 3-D models to mimic elements of a previously designed mod-kit drawings and modify spacial query functions to consider the instructions when computing 3-D space.
- Supported the effort to migrate programs from outdated Catia V4 systems into our Catia V5 system for drawing release and 3-D design. Specifically, supported processes that made this an easy transition for legacy programs.

Skills

AI/ML skills: Numpy, Tensorflow, pyTorch, OpenCV, DepthAI, TensorRT, SNPE, Matplotlib, Image Processing
Programming Skills: C++, C#, python, Catia/3dx CAA, ROS, bash, powershell, GitLab CI/CD
Hardware: Luxonis, Stereolabs ZED X, Ouster Lidar, Nvidia Jetson, Qualcomm

Personal Projects

LearnedRubiks **GitHub**
iOS app that scans a cube with the camera, guides the solve, and teaches each stage Skills: Swift, iOS

Euchre-like Roguelike **GitHub**
Roguelike take on Euchre, a trump trick-taking game; upgrade your deck between rounds Skills: Godot, C#